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Inventor(s)	Nagarajan; Radhakrishnan L.

Integrated coherent optical transceiver

Abstract

An integrated circuit includes a silicon photonics substrate and includes a transimpedance amplifier (TIA) chip and a driver chip arranged on the silicon photonics substrate. The silicon photonics substrate includes a silicon-based material, includes electrical connections, and includes silicon photonics components configured to receive and transmit optical signals. The TIA chip includes a silicon-germanium material that is different from the silicon-based material, is connected via the electrical connections to at least one of the silicon photonics components configured to receive an optical signal, and is configured to process a received optical signal and to output a processed signal to a digital signal processor. The driver chip includes CMOS material that is different from the silicon-germanium material and the silicon-based material, and is connected via the electrical connections to drive at least one of the silicon photonics components configured to generate an optical signal for transmission.

Inventors:	Nagarajan; Radhakrishnan L. (Santa Clara, CA)
Applicant:	Marvell Asia Pte Ltd (Singapore, SG)
Family ID:	1000008818518
Assignee:	MARVELL ASIA PTE LTD (Singapore, SG)
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Primary Examiner: Wong; Eric

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is continuation of and claims priority to U.S. patent application Ser. No. 17/867,567, filed on Jul. 18, 2022 (now issued as U.S. Pat. No. 11,811,433, issued Nov. 7, 2023), which is a continuation of U.S. patent application Ser. No. 16/928,845, filed on Jul. 14, 2020 (now issued as U.S. Pat. No. 11,394,464, issued), which is a continuation of U.S. patent application Ser. No. 16/357,095, filed on Mar. 18, 2019, (now issued as U.S. Pat. No. 10,754,091 on Aug. 25, 2020), commonly assigned and incorporated by reference herein for all purposes.

BACKGROUND OF THE INVENTION

(1) The present invention relates to optical telecommunication techniques. More particularly, the present invention provides a compact integrated coherent transceiver based on silicon photonics platform, a method for forming the same, and a system having the same.

(2) Over the last few decades, the use of communication networks exploded. In the early days Internet, popular applications were limited to emails, bulletin board, and mostly informational and text-based web page surfing, and the amount of data transferred was usually relatively small. Today, Internet and mobile applications demand a huge amount of bandwidth for transferring photo, video, music, and other multimedia files. For example, a social network like Facebook processes more than 500 TB of data daily. With such high demands on data and data transfer, existing data communication systems need to be improved to address these needs.

(3) 40-Gbit/s and then 100-Gbit/s data rates wide-band DWDM (Dense Wavelength Division Multiplexed) optical transmission over existing single-mode fiber is a target for the next generation of fiber-optic communication networks. Chip-scale wide-band wavelength-tunable lasers have been of interest for many applications such as wide-band DWDM communication and wavelength-steered light detection and ranging (LIDAR) sensing. More recently, optical components are being integrated on silicon (Si) substrates for fabricating large-scale photonic integrated circuits that co-exist with micro-electronic chips. A whole range of photonic components, including filters, (de)multiplexers, splitters, modulators, and photodetectors, have been demonstrated, mostly in the silicon-on-insulator (SOI) platform. The SOI platform is especially suited for standard WDM communication bands at 1300 nm and 1550 nm, as silicon ($n=3.48$) and its oxide SiO_2 ($n=1.44$) are both transparent, and form high-index contrast, high-confinement waveguides ideally suited for medium to high-integration planar integrated circuits (PICs).

(4) Coherent optical fiber communications were studied extensively in the 1980s mainly because high sensitivity of coherent receivers could elongate the unrepeated transmission distance; however, their research and development have been interrupted for nearly 20 years behind the rapid progress in high-capacity wavelength-division multiplexed (WDM) systems using erbium-doped fiber amplifiers (EDFAs). Not long ago, the demonstration of digital carrier phase estimation in coherent receivers has stimulated a widespread interest in coherent optical communications again. The fact is, the digital coherent receiver enables us to employ a variety of spectrally efficient modulation formats such as phase shift keying (PSK) and quadrature amplitude modulation (QAM) without relying upon a rather complicated optical phase-locked loop. In addition, since the phase information is preserved after detection, we can realize electrical post-processing functions such as compensation for chromatic dispersion and polarization-mode dispersion in the digital domain. These advantages of the coherent receiver have enormous potential for innovating existing optical communication systems.

(5) Coherent transmitter has a TE and TM path. Silicon photonics chip, however, operates essentially only in a TE only configuration. Several technical challenges exist for making polarization-independent or wavelength tunable passive and active components as well as integrating these components to form a coherent optical transceiver in a compact silicon photonics platform. Therefore, improved techniques and methods of forming an integrated compact coherent transceiver are desired.

BRIEF SUMMARY OF THE INVENTION

(6) The present invention relates to optical telecommunication techniques. More particularly, the present invention provides an integrated compact coherent transceiver in silicon-photonics platform. Merely by example, the present invention discloses a coherent transmitter block including a polarization independent semiconductor optical amplifier (SOA) coupled via polarization beam rotator combiner (PBRC) to a driver electronics chip with both in-phase and quadrature-phase modulations of broadband signals from an independently packaged wide-band tunable laser with

silicon-photonics tuning section, and a coherent receiver block including polarized hybrid receivers converting TE-polarized optical signals split by a polarization beam splitter rotator (PBSR) to electrical signals for a transimpedance amplifier (TIA) electronics chip, and a method of integrating these components to form a coherent transceiver in a compact silicon photonics platform for wide-band DWDM optical communications, though other applications are possible.

(7) In modern electrical interconnect systems, high-speed serial links have replaced parallel data buses, and serial link speed is rapidly increasing due to the evolution of CMOS technology. Internet bandwidth doubles almost every two years following Moore's Law. But Moore's Law is coming to an end in the next decade. Standard CMOS silicon transistors will stop scaling around 5 nm. And the internet bandwidth increasing due to process scaling will plateau. But Internet and mobile applications continuously demand a huge amount of bandwidth for transferring photo, video, music, and other multimedia files. This disclosure describes techniques and methods to improve the communication bandwidth beyond Moore's law.

(8) With the increase in the transmission capacity of WDM systems, coherent technologies attracted a renewal of widespread interest after 2000. The motivation is to develop methods for meeting the ever-increasing bandwidth demand with multi-level modulation formats based on coherent technologies. The first step in the revival of coherent optical communications research was triggered by quadrature phase-shift keying (QPSK) modulation/demodulation experiment featuring optical in-phase and quadrature (IQ) modulation (IQM) and optical delay detection. In such a scheme, we can double the bit rate while maintaining the symbol rate, and a 40-Gb/s differential QPSK (DQPSK) system has been put into practical use. Coherent transmitter has a transverse electric (TE) mode and transverse magnetic (TM) mode path. Silicon photonics chip operates essentially only in a TE only configuration. The signal has both TE mode and TM mode parts. To accomplish a silicon-photonics based integration of coherent transceiver, we need to have at least the following two components on the silicon photonics chip: 1) Polarization Beam Rotator Combiner (PBRC) integrated with a polarization-independent semiconductor optical amplifier (SOA) and driver in the coherent transmitter and 2) Polarization Beam Splitter Rotator (PBSR) integrated with electronics like transimpedance amplifier (TIA) in the coherent receiver, and to integrate the coherent transmitter and receiver with a tunable laser provided in a flip-chip.

(9) In an embodiment, the present invention provides a tunable laser device based on silicon photonics. The tunable laser device includes a substrate configured with a patterned region comprising one or more vertical stoppers, an edge stopper facing a first direction, a first alignment feature structure formed in the patterned region along the first direction, and a bond pad disposed between the vertical stoppers. The tunable laser device further includes an integrated coupler built in the substrate located at the edge stopper. Additionally, the tunable laser device includes a laser diode chip including a gain region covered by a P-type electrode and a second alignment feature structure formed beyond the P-type electrode, the laser diode chip being flipped to rest against the one or more vertical stoppers with the P-type electrode attached to the bond pad and the gain region coupled to the integrated coupler. Furthermore, the tunable laser device includes a tuning filter fabricated in the substrate and coupled via a wire waveguide to the integrated coupler.

(10) In a specific embodiment, the present invention provides a coherent transceiver integrated on silicon photonics substrate. The coherent transceiver includes a substrate member and a tunable laser device comprising a laser diode chip having a gain region with a p-side electrode flipped down and mounted on the substrate member. The gain region is coupled with a wavelength tuning section formed in the substrate member to tune wavelengths of a laser light outputted from the gain region to a waveguide in the substrate member. The coherent transceiver further includes a first power splitter coupled to the waveguide to split the laser light to a first light and a second light. Additionally, the coherent transceiver includes a coherent receiver block including at least two 90° hybrid receivers coupled respectively to two outputs of a polarization beam splitter rotator in the substrate member to receive a coherent input signal from a coherent optical network. The coherent

transceiver also is configured to have the two 90° hybrid receivers to couple with two outputs of a second power splitter to receive two local-oscillator signals split from the first light for assisting detections of a transverse electric (TE) mode signal and a transverse magnetic (TM) mode signal in the coherent input signal. Furthermore, the coherent transceiver includes a coherent transmitter block including at least a pair of in-phase/quadrature-phase modulators in the substrate member to respectively modulate two parts split from the second light to two I/Q-modulated signals in TE-mode, a polarization beam rotator combiner in the substrate member to rotate one of the two I/Q modulated signals to a TM-mode signal and combine with the other one of two I/Q modulated signals in TE-mode to generate a coherent output signal transmitted through a polarization-independent semiconductor optical amplifier to the coherent optical network.

(11) In another specific embodiment, the present invention provides a module package of an integrated coherent optical transceiver. The module package includes a metal case having two side members joint by a joint member coupled with a bottom member and clipped with a lid member. The module package further includes a printed circuit board (PCB) disposed on the bottom member with multiple electrical pins located near a back end of the metal case and a silicon photonics substrate with a board grid array at bottom mounted on the PCB. Additionally, the module package includes a coherent transceiver chip integrated on the silicon photonics substrate including one input port and one output port respectively disposed on a front end of the metal case and coupled to the silicon photonics substrate via a first fiber and a second fiber. The coherent transceiver chip includes further a tunable laser device including two laser diode chips coupled with a wavelength tuning section embedded in a first region of the silicon photonics substrate and configured to output a laser light. Furthermore, the coherent transceiver chip includes a transimpedance amplifier (TIA) chip flip-mounted on a second region of the silicon photonics substrate and a driver chip flip-mounted on a third region of the silicon photonics substrate. Moreover, the coherent transceiver chip includes a silicon photonics circuit formed in a fourth region of the silicon photonics substrate. The silicon photonics circuit is configured to couple with the first fiber to receive a coherent input light signal and to couple with the tunable laser device to receive a first portion of a laser light as local oscillator signals to assist detections of both TM-mode and TE-mode light signals in the coherent input light signal by the TIA chip. The silicon photonics circuit is also configured to use the driver chip to drive modulations of a second portion of the laser light to generate a coherent output light signal outputted to the second fiber.

(12) In another specific embodiment, the present invention provides a light engine device including an optical coherent transceiver integrated on a semiconductor substrate member. The light engine device includes a substrate member comprising a surface region. The light engine device further includes an optical input configured to an incoming fiber device and an optical output configured to an outgoing fiber device. Additionally, the light engine device includes a transmit path provided on the surface region. The transmit path includes a polarization independent optical amplifier device coupled to the optical output. The transmit path also includes a polarization beam rotator combiner device coupled to the polarization independent optical amplifier and coupled to the optical output. The transmit path further includes a dual polarization I/Q Mach Zehnder modulator device coupled to the polarization beam rotator combiner device and coupled to the optical output. The transmit path again includes a driver device coupled to the dual polarization I/Q Mach Zehnder modulator device and configured to drive an electrical signal to the dual polarization I/Q Mach Zehnder modulator. Furthermore, the transmit path includes a tunable laser device comprising a laser diode chip having a gain region with a p-side electrode flipped down and mounted on the substrate member. The gain region is coupled with a wavelength tuning section formed in the substrate member to tune wavelengths of a laser light outputted from the gain region to a waveguide in the substrate member. Moreover, the transmit path includes a first power splitter coupled to the waveguide to split the laser light to a first light and a second light. The second light is coupled to the dual polarization I/Q Mach Zehnder modulator device. Further, the light engine device also

includes a receive path provided on the surface region. The receive path includes a second power splitter coupled to the first light. The receive path further includes a pair of 90° hybrid receivers. Each of the pair of 90° hybrid receivers includes a photo detector device and a hybrid mixer device, coupled respectively to two outputs of a polarization beam splitter rotator in the substrate member to receive the optical input and to two outputs of the second power splitter to receive the first light from the tunable laser device for assisting detections of a transverse electric (TE) mode signal and a transverse magnetic (TM) mode signal in the coherent input signal. Furthermore, the receive path includes a transimpedance amplifier coupled to each of the 90° hybrid receivers and coupled to each of the photo detector devices that convert a combination of the first light with the optical input into an electrical signal to be transmitted to using the transimpedance amplifier device. The light engine device further includes a heterogeneous integration configured using the substrate member, the transmit path, and the receive path to form a single silicon photonics device.

(13) The present invention achieves these benefits and others in the context of known waveguide laser modulation technology. However, a further understanding of the nature and advantages of the present invention may be realized by reference to the latter portions of the specification and attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following diagrams are merely examples, which should not unduly limit the scope of the claims herein. One of ordinary skill in the art would recognize many other variations, modifications, and alternatives. It is also understood that the examples and embodiments described herein are for illustrative purposes only and that various modifications or changes in light thereof will be suggested to persons skilled in the art and are to be included within the spirit and purview of this process and scope of the appended claims.

(2) FIG. 1 is a simplified block diagram of a coherent optical transceiver according to an embodiment of the present invention.

(3) FIG. 2A is a top-view diagram of a waveguide-based polarization beam splitter-rotator (PBSR) or polarization beam rotator-combiner (PBRC) according to an embodiment of the present invention.

(4) FIG. 2B is a cross-section view along AA' plane of the waveguide-based PBSR or PBRC of FIG. 2A according to an embodiment of the present invention.

(5) FIG. 2C is a cross-section view along BB' plane of the waveguide-based PBSR or PBRC of FIG. 2A according to an embodiment of the present invention.

(6) FIG. 2D is a cross-section view along CC' plane of the waveguide-based PBSR or PBRC of FIG. 2A according to an embodiment of the present invention.

(7) FIG. 3 is a simplified diagram of a silicon photonics tunable laser device according to an embodiment of the present invention.

(8) FIG. 4 is a schematic diagram showing a perspective view a laser diode chip flipped bonding to a silicon photonics substrate according to an embodiment of the present invention.

(9) FIG. 5 is an exemplary diagram of a wavelength tuning map of a silicon photonics tunable laser according to an embodiment of the present invention.

(10) FIG. 6 is a simplified diagram of a silicon photonics tunable laser device according to another embodiment of the present invention.

(11) FIG. 7 is a flowchart of a method for tuning wavelength of a laser output of the silicon photonics tunable laser device according to an embodiment of the present invention.

(12) FIG. 8 is an exemplary diagram of two superimposed transmission spectra of respective two ring resonators with different radii of a tunable filter in the silicon photonics tunable laser device

according to an embodiment of the present invention.

(13) FIG. 9 is an exemplary diagram of a reflectivity spectrum of a reflector coupled to the two ring resonators of the tunable filter according to the embodiment of the present invention.

(14) FIG. 10 is a simplified block diagram of the tunable filter including two ring resonators, a reflector, plus a phase shifter in a Vernier ring reflector configuration according to an embodiment of the present invention.

(15) FIG. 11 is an exemplary diagram of two synthesized spectra respectively corresponding to wavelengths being tuned from 1555 nm to 1535 nm by tuning the tunable filter in Vernier ring reflector configuration and corresponding gain profile from 1530 nm to 1570 nm according to the embodiment of the present invention.

(16) FIG. 12 is an exemplary diagram of laser spectra outputted by the silicon photonics tunable laser device with laser wavelength being tuned from 1555 nm to 1535 nm according to an embodiment of the present invention.

(17) FIG. 13 is a schematic diagram of three types of an integrated coupler based on SiN in Si waveguide according to some embodiments of the present invention.

(18) FIG. 14A is an exemplary diagram of a relationship between coupling loss and a lateral misalignment for an integrated coupler coupled between the tunable filter and a laser diode chip according to an embodiment of the present invention.

(19) FIG. 14B is an exemplary diagram of a relationship between coupling loss and a vertical misalignment for the integrated coupler coupled between the tunable filter and a laser diode chip according to the embodiment of the present invention.

(20) FIG. 15A is a perspective view of an integrated coherent optical transceiver on a silicon photonics substrate according to an embodiment of the present invention.

(21) FIG. 15B is a side view of the integrated coherent optical transceiver on a silicon photonics substrate of FIG. 15A according to an embodiment of the present invention.

(22) FIG. 16A is a schematic diagram of an open package of an integrated coherent optical transceiver according to an embodiment of the present invention.

(23) FIG. 16B is a schematic diagram of a closed package of the integrated coherent optical transceiver of FIG. 16A according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(24) The present invention relates to optical telecommunication techniques. More particularly, the present invention provides an integrated compact coherent transceiver in silicon photonics platform. Merely by example, the present invention discloses a coherent transmitter block including a polarization independent semiconductor optical amplifier (SOA) coupled via polarization beam rotator combiner (PBRC) to a driver electronics chip and to an independently packaged wide-band tunable laser with silicon photonics tuning section, and a coherent receiver block including polarized hybrid receivers coupled to transimpedance amplifier (TIA) electronics chip via polarization beam splitter rotator (PBSR), and a method of integrating these components to form a coherent transceiver in a compact silicon photonics platform for wide-band DWDM optical communications, though other applications are possible.

(25) The following description is presented to enable one of ordinary skill in the art to make and use the invention and to incorporate it in the context of particular applications. Various modifications, as well as a variety of uses in different applications will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to a wide range of embodiments. Thus, the present invention is not intended to be limited to the embodiments presented, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

(26) In the following detailed description, numerous specific details are set forth in order to provide a more thorough understanding of the present invention. However, it will be apparent to one skilled in the art that the present invention may be practiced without necessarily being limited to these

specific details. In other instances, well-known structures and devices are shown in block diagram form, rather than in detail, in order to avoid obscuring the present invention.

(27) The reader's attention is directed to all papers and documents which are filed concurrently with this specification and which are open to public inspection with this specification, and the contents of all such papers and documents are incorporated herein by reference. All the features disclosed in this specification, (including any accompanying claims, abstract, and drawings) may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

(28) Furthermore, any element in a claim that does not explicitly state “means for” performing a specified function, or “step for” performing a specific function, is not to be interpreted as a “means” or “step” clause as specified in 35 U.S.C. Section 112, Paragraph 6. In particular, the use of “step of” or “act of” in the Claims herein is not intended to invoke the provisions of 35 U.S.C. 112, Paragraph 6.

(29) Please note, if used, the labels left, right, front, back, top, bottom, forward, reverse, clockwise and counter clockwise have been used for convenience purposes only and are not intended to imply any particular fixed direction. Instead, they are used to reflect relative locations and/or directions between various portions of an object.

(30) In an aspect, the present disclosure provides a compact integrated coherent transceiver based on silicon photonics platform. As data transmission-capacity increase in WDM systems, coherent technologies with polarized optical signals also increasingly attract more and more interest over the recent years with motivation of meeting the ever-increasing bandwidth demand using multilevel modulation formats. For example, a digital signal transmitting/receiving scheme is developed for coherent transmission system supported with a quadrature PSK (QPSK) modulation/demodulation featuring optical in-phase and quadrature (I/Q) modulation and optical delay detection. In such a scheme, one symbol carries two bits by using the four-point constellation on the complex plane, therefore, the bit rate is doubled, while keeping the symbol rate, or maintaining the bit rate even with the halved spectral width to realize a large capacity of 100 Gbit/s and more than 100 Gbit/s per channel. Optical coherent I/Q modulation can be realized with Mach-Zehnder (MZ) type push-pull modulators in parallel, between which a $\pi/2$ -phase shift is given. The I/Q components of the optical carrier is modulated independently with the coherent I/Q modulator, enabling any kind of modulation formats.

(31) FIG. 1 shows simplified block diagram of an integrated coherent optical transceiver according to an embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, the integrated coherent optical transceiver is configured to integrate a silicon-photonics-based tunable laser **1000** with a coherent transmitter block **2000** and a coherent receiver block **3000** on a single silicon photonics substrate **100**.

Optionally, the tunable laser is a wide-band laser source with its output optical signal tunable over entire C-band or O-band. The wavelength tuning is achieved using a silicon photonics tuning section including a heater-tuned micro-ring resonator filter built in the silicon photonics substrate **100** while a laser diode chip with an InP gain region with p-side flip down mounted onto the silicon photonics substrate **100**. Particularly, through the silicon photonics tuning section and a waveguide-based wavelength locker, laser wavelength can be tuned to provide all ITU channels in C-band each with a sharp transmission spectrum for DWDM communication. More details can be found descriptions of FIG. 3 through FIG. 14 of the specification. Optionally, the laser light is outputted from the InP gain region coupled via an integrated coupler with a waveguide (formed in a same silicon photonics substrate **100**). The waveguide delivers the laser light, which is used as a local oscillator source for the coherent receiver **3000** to detect incoming coherent light signals and is also served as a light source for producing I/Q modulated coherent signal to be transmitted by the

coherent transmitter **2000**.

(32) Optionally, the integrated coherent optical transceiver includes multiple silicon waveguides respectively laid in the silicon photonics substrate **100**, though they are not explicitly labeled in FIG. **1**, for connecting several different silicon photonics components including power splitter, polarization beam splitter and rotator, polarization beam rotator and combiner, phase shifter, power attenuator, or part of optical modulator. Optionally, the silicon waveguides have regular rectangular wire shape with a fixed width and height. Optionally, the height is selected based on a usage of standard 220 nm silicon-on-insulator (SOI) substrate during its formation process. Optionally, the width of the silicon waveguide is about 300 nm to 500 nm. Optionally, the silicon waveguides have alternative shaped structures like rib structure with multiple steps in height, taper structure with varying widths along its length, or multiple branches of different widths and separations joined at different cross-section planes, depending on specific functional applications. Optionally, some of the silicon photonics components mentioned above are also silicon waveguides themselves monolithically formed in a same manufacture process for preparing the silicon photonics substrate **100** to integrate the coherent optical transceiver.

(33) For ultrasmall silicon waveguides of a few hundred of nanometers in width (parallel to the substrate) and height (orthogonal to the substrate) used in compact silicon photonics modules, they often exhibit very strong polarization dependence for a transmitted light with infrared wavelength (typical for telecommunications) such that the transmission loss for a transverse electric (TE) mode polarization (parallel to the substrate) is much less than that for a transverse magnetic (TM) mode polarization (orthogonal to the substrate). Therefore, typical silicon photonics device works for TE mode only. Optionally, the output light of the tunable laser **1000** in a transverse electric (TE) mode polarization is split between the transmitter block **2000** and the receiver block **3000** with an optimized ratio. In the transmitter block **3000**, the light coming from the tunable laser **1000** at a wavelength selected in a broadband by a tunable filter section is firstly split to a TE-mode light and a TM*-mode (still a TE-mode) light, both of which are performed an optical I/Q modulation. The TE-mode light becomes a TE-mode signal. The TM*-mode light is then performed a polarization rotation to form a transverse magnetic (TM) mode signal. The TE-mode signal and the TM-mode signal are then combined to become a coherent output signal of the transceiver at the selected wavelength. In the receiver block **3000**, the optical signal (TE-mode) coming from the tunable laser **1000** acts as a local oscillator at the wavelength selected by the tunable filter section for detecting incoming coherent input signals.

(34) Referring to FIG. **1**, the output of the tunable laser **1000** is substantially a TE-mode laser light, which is firstly split by a power splitter **1001** to two parts, a first light and a second light respectively with a X:Y ratio. The first light is delivered to the coherent receiver block **3000** with X portion of the laser power and the second light is forwarded to the coherent transmitter block **2000** with Y portion of the laser light. The X:Y ratio is varied or optimized based on specific system operation condition. Optionally, X is set in a range from 10 to 50%. For the first light coupled into the coherent receiver block **3000**, it is further split to two 50% portions, a third light and a fourth light, by a power splitter **3001**. The third light is loaded into a TE 90° hybrid receiver **3200** and the fourth light is loaded into a TM* 90° hybrid receiver **3300**, for respectively assisting detections of TE-mode/TM-mode signals of the incoming optical signal received by the coherent optical transceiver via an input port (Light in).

(35) For the second light coupled into the coherent transmitter block **2000**, it is used as an original light source to generate a coherent optical signal to be transmitted out of an output port. Optionally, the second light can be provided as a TE-mode polarized light and tuned to be at any wavelength in a broadband like C-band or O-band designated for telecommunication. In order to serve as a coherent light signal with mixed polarization modes, a TM-mode polarized part of the second light needs to be generated. Both TE- and TM-mode polarized parts are modulated by optical coherent modulators. Optionally, the optical coherent I/Q modulator based on delay-line interferometer in

silicon or silicon nitride or similar material integrated in silicon photonics substrate has been developed with polarization compensation at selective wavelengths, e.g., as shown in U.S. Pat. No. 10,031,289, assigned to Inphi Corp.

(36) As seen in FIG. 1, both the coherent transmitter block **2000** and the coherent receiver **3000** of the integrated coherent optical transceiver have a TE-mode path and a TM-mode path for providing or detecting a coherent light with mixed polarizations in the polarization-independent communication system. However, silicon photonics circuit including Si-based waveguide linear modulator operates essentially in a TE-mode only configuration. In order to handle the coherent optical signal with both TE mode and TM mode parts, the silicon-photonics based coherent transceiver needs at least 1) a Polarization Beam Rotator Combiner (PBRC) integrated with a polarization-independent semiconductor optical amplifier (SOA) at an output port (Light out) of the transmitter block to output a coherent optical signal having both TE and TM mode, and 2) a Polarization Beam Splitter Rotator (PBSR) to split TE-mode part and TM-mode part of the incoming coherent optical signal and rotate the TM-mode part to TE-mode before being coupled into optical hybrid detectors.

(37) Conventional PBSR is either wavelength sensitive so that it is not suit for broadband operation or based on prism that is hard to be made in super compact size. A compact PBSR based on photonic integrated circuits (PICs) having combined features like compact size, high extinction ratio, low insertion loss, broadband range, stability, simple structure and high tolerances in manufacture has been developed, referring to U.S. Pat. No. 9,915,781 assigned to Inphi Corp. In particular, as shown in FIG. 2A, a waveguide-based polarization beam splitter-rotator is shown in its top view. The PBSR **16** includes a single piece of monolithically silicon planar waveguide including multiple shaped sections formed with a standard 220 nm silicon layer of a silicon-on-insulator (SOI) substrate. A first section is a converter in rib structure characterized, in both FIG. 2A and FIG. 2B, by a top-layer **1602** overlying a bottom-layer **1601** extended in the lengthwise direction through a first segment of a length $L_{sub.1}$ from the input port **1600** to a joint plane AA' and a second segment of a length $L_{sub.2}$ from the joint plane to the first cross-section plane **1610**. The top-layer **1602** is narrower than the bottom-layer **1601** and both vary throughout the first length $L_{sub.1}$ and throughout the second length $L_{sub.2}$ except they have a first common width $W_{sub.0}$ at the input port **1600** and a second common width $W_{sub.1}$ at the first cross-section plane **1610**. The specific length-width combination of both the top-layer **102** and the bottom-layer **1601** is configured to provide a polarization-mode conversion of TM-mode to TE-mode for a coherent light wave transmitted through, depending on wavelength ranges of the coherent optical signal.

(38) The top-layer **1602** of a thickness of h_t is formed overlying the bottom-layer **1601** of a thickness $h_{sub.b}$ in an overlay process after the silicon layer of the thickness of $h = h_{sub.t} + h_{sub.b} = 220$ nm over an oxide layer is patterned for the rib structure waveguide as part of a monolithic process of forming the PBSR **16**. In a specific embodiment, the converter is configured for handling optical wave of broadband wavelengths, for example, O-band in a range of 1270 nm to 1330 nm or C-band from about 1530 nm to about 1560 nm (with slightly different dimensions of the rib structure). In addition, the width $W_{sub.t}$ of the top-layer **1602** at the joint plane is made to be greater than the first common width $W_{sub.0}$, the width $W_{sub.b}$ of the bottom-layer **1601** at the joint plane is made to be greater than the width $W_{sub.t}$ of the top-layer **1602** but smaller than the second common width $W_{sub.1}$, and the first length $L_{sub.1}$ is made to be shorter than the second length $L_{sub.2}$. After fine tuning the length-width combination (with a standard height of 220 nm) under the above configuration the rib structure waveguide serves a desired polarization mode converter. For an input light with mixed TM mode and TE mode inputted via the input port **1600**, the TM mode is substantially converted to first-order Transverse Electric (TE₁) mode and the TE mode is substantially converted to zero-order Transverse Electric (TE₀) mode as the input light travels to the first cross-section plane **1610**. Specifically, the TE₁ mode includes two sub-modes, an out-of-phase TE_{1,sub.1} sub-mode and an in-phase TE₁₂ sub-mode. The TE₀ mode

just is a single-phase mode.

(39) Referring to FIG. 2A, a second shaped section of the PBSR **16** includes a splitter **1612** directly coupled to the first cross-section plane of the converter as part of the monolithic planar silicon waveguide formed from the 220 nm silicon layer of the SOI substrate. FIG. 2C is a cross-section view of the waveguide-based PBSR **10** along BB' plane according to an embodiment of the present invention. Referring to FIG. 2C and FIG. 2A, the splitter **1612** is a planar waveguide having a height h of the 220 nm silicon layer extended in the lengthwise direction from the first cross-section plane **1610** to a second cross-section plane **1620**. The first cross-section plane **1610** passes the input light transmitted through the converter. The second cross-section plane **1620** includes a first port **1701** and a second port **1702** respectively located next to two opposing edges and separated from each other by a gap $W_{sub.g}$. In an embodiment, the splitter **1612** is designed for splitting the input light received at the first cross-section plane **1610** substantially evenly to a first wave at the first port **1701** and a second wave at the second port **1702**.

(40) Referring to FIG. 2A again, a third shaped section of the PBSR **16** includes a phase shifter waveguide coupled to or naturally extended from the first port **1701** and the second port **1702** at the second middle cross-section plane **1620**. The phase shifter waveguide includes a first waveguide arm **1621** coupled to the first port **1701** and a second waveguide arm **1622** coupled to the second port **1702**, both having the same height h of the 220 nm silicon layer. The first waveguide arm **1621** is extended in the lengthwise direction to a third port **1801** of a third cross-section plane **1630** and the second waveguide arm **1622** is separately extended in the lengthwise direction to a fourth port **1802** of the third cross-section plane **1630**. In an embodiment, the first waveguide arm **1621** is configured to receive the first wave from the first port **1701** and to transmit the first wave through at least a length $L_{sub.6}$ towards the third port **1801** while keeping the first wave at the third port **1801** in-phase relative to that at the first port **1701**. The second waveguide arm **1622** is configured to receive the second wave from the second port **1702** and to transmit the second wave through a separate path of the same length $L_{sub.6}$ towards the fourth port **1802** while adding a phase shift to the second wave at the fourth port **1802** relative to that at the second port **1702**.

(41) FIG. 2D is a cross-section view of the waveguide-based PBSR **16** of FIG. 2A along CC' plane according to an embodiment of the present invention. Referring to FIG. 2A and FIG. 2D, the first waveguide arm **1621** of the phase shifter includes a straight bar shape of at least the length $L_{sub.6}$ and a first arm width $W_{sub.1a}$ connected between the first port **1701** and the third port **1801**, and the second waveguide arm **1622** of the phase shifter includes a straight bar shaped portion having at least the length $L_{sub.6}$ joined aside with a triangle shaped portion connected between the second port **1702** and the fourth port **1802**. The second waveguide arm **1622** has a varying second arm width $W_{sub.2a}$ which increases from the first arm width $W_{sub.1a}$ at one end to a maximum at an apex of the triangle shaped portion then decreasing again to the first arm width $W_{sub.1a}$ at the other end. The constant width $W_{sub.1a}$ associated with the length $L_{sub.6}$ in the first waveguide arm **1621** effectively retains first wave in-phase travelling through the first waveguide arm **1621** to reach the third port **1801** at the third cross-section plane **1630**. At the same time, the varying width $W_{sub.2a}$ associated with the length $L_{sub.6}$ in the second waveguide arm **1622** can be adjusted to provide a desired phase delay to the second wave traveling independently through the second waveguide arm **1622** to reach the fourth port **1802** at the third cross-section plane **1630**. In a specific configuration, the maximum $W_{sub.2a}$ is set to be slightly smaller than twice of the first arm width $W_{sub.1a}$ and the length $L_{sub.6}$ is no greater than $11\ \mu\text{m}$ to cause a phase delay of $(\frac{1}{2})\pi$ in the second wave through the second waveguide arm **1622**. Alternatively, with lightly reduction in the maximum $W_{sub.2a}$ and increase in the length $L_{sub.6}$, a phase delay of $(\frac{3}{2})\pi$ can be produced to the second wave through the second waveguide arm **1622**. In principle, a phase shift of $(\pi/2 + n\pi)$ can be generated for n being any integer though effective phase values are all limited within 2π .

(42) Referring to FIG. 2A again, a fourth shaped section of the PBSR **16** includes a 2×2 MMI

coupler **1632** as a planar waveguide of the same height h of 220 nm silicon layer naturally extended from the third port **1801** and the fourth port **1802** at the third cross-section plane **1630** to an output plane **1640** with a first output port **1901** and a second output port **1902**. From the input port **1600** to the fourth cross-section plane **1640**, the PBSR **16** includes a total length less than 100 μm , thus forming a very compact sized device suitable for highly integrated silicon photonics communication system. The 2×2 MMI coupler **1632** is characterized by a rectangular shape of a length $L_{\text{sub.7}}$ measured from the third cross-section plane **1630** to the output plane **1640** and a width $W_{\text{sub.2}}$. The first output port **1401** is aligned with the third port **1801** in a bar position at a distance $W_{\text{sub.p}}$ away from a central line of the rectangular shaped planar waveguide in the lengthwise direction. The first output port **1901** is in a cross position relative to the fourth port **1802**. The second output port **1902** and the fourth port **1802** are respectively in mirror symmetric positions relative to the first output port **1901** and the third port **1801**, nevertheless making the second output port **1902** to be in a cross-position relative to the third port **1801**. The 2×2 MMI coupler **1632** in such configuration induces a general interference of optical waves coupled via both the third port **1801** and the fourth port **1802** and outputs a first output light in TE₀ mode to the first output port **1901** and a second output light in TE₀ mode to the second output port **1902**. Depending on specific polarization modes and phase difference of the first wave at the third port **1801** and the second wave at the fourth port **1802**, optionally, the first output light may be exclusively originated from the input light with TM mode and the second output light may be exclusively originated from the input light with TE mode. In other words, the PBSR **16**, with a compact length less than 100 μm , is able to split the input light with mixed TM mode part and TE mode part, and to guide substantially TM mode part and rotate it to a TE₀ mode outputted to a first output port, and guide substantially TE mode part and retain it as TE mode outputted to a second output port, realizing a polarization splitting/rotating function. And this function is applicable to broadband transmission with any wavelength in entire O-band or C-band.

(43) Referring back to FIG. **1**, for the coherent receiver block **3000** the incoming coherent light signal R has mixed TE/TM-mode parts in I/Q modulation. A PBSR **3100** is used to receive the incoming light signal R and, following the polarization splitting/rotating function described above, to output the TE-mode part signal $R_{\text{sub.E}}$ to a first waveguide in TE polarization and the TM-mode part R_{M} to a TM*-branch also in TE polarization, compatible for the silicon photonics chip that operates in the TE-mode only. The TE-mode part signal $R_{\text{sub.E}}$ in the TE-branch is then loaded into a TE 90° hybrid receiver **3200** which also receives a local-oscillation (LO) signal A in TE-mode split from the tunable laser **1000** as a second input (FIG. **1**). The TE 90° hybrid receiver **3200** produces four outputs: $\frac{1}{2}\times(R_{\text{sub.E}}+A)$, $\frac{1}{2}\times(R_{\text{sub.E}}-A)$, $\frac{1}{2}\times(R_{\text{sub.E}}+jA)$, and $\frac{1}{2}\times(R_{\text{sub.E}}-jA)$ through a perfect I/Q demodulation, into a trans-impedance amplifier (TIA) chip **3400** where these optical signals are converted to two current signals I and Q by balanced photodetectors. Further, the TIA chip includes analog-to-digital converter to convert the current signals into digital signals that can be processed in a digital signal processor (DSP) (not explicitly shown) and the TE-mode part signal of the incoming coherent light can be specifically detected. Similarly, the TE-mode part signal R_{M} in the TM*-branch is loaded into a TM* 90° hybrid receiver **3300** which is also receives a LO signal A in TE-mode split from the tunable laser **1000** as a second input. Also, the outputs of the TM* 90° hybrid receiver **3300** are sent to the TIA chip **3400** where the TM-mode part signal of the incoming coherent light can be specifically detected via I/Q demodulation. Optionally, the TIA chip **3400** is a flip-chip mounted on the silicon photonics substrate **100** shared with the silicon photonics chip including the PBSR **3100** and power splitter **3001**.

(44) Referring to FIG. **1** again, the coherent transmitter block **2000** receives the Y portion light in TE-mode split from the tunable laser **1000** which is non-modulated and needs optical I/Q modulation for coherent communications. In the embodiment, the coherent transmitter block **2000** includes a driver chip **2600** configured to provide bias voltages and digital transmit electrical

signals for an optical modulator to modulate the input light from the tunable laser **1000**. The driver chip **2600** (e.g., including a serializer, a 4-level encoder, a 2-bit digital-to-analog converter) is provided in a flip-chip mounted on the same silicon photonics substrate **100**. Optionally, the driver chip **2600** and the TIA chip **3400** can be integrated into a single chip. In the embodiment, a coherent I/Q modulator based on delay-line interferometer made by silicon or silicon nitride or similar material integrated in the silicon photonics chip that shares the same silicon photonics substrate **100** with the coherent receiver block **3000**. The Y portion light in TE-mode from the tunable laser **1000** is split to a first input light in a TE Branch **2501** and a second input light in a TM* Branch **2502**. Here the split of the Y portion light is done via a 50:50 power splitter (not explicitly shown) while retaining its TE polarization in either branch. Each of the first input light and the second input light is coupled into an I/Q modulator. In the embodiment, a first I/Q modulator is configured to have four linear waveguide arms with different phase delays to receive four equally split portions of the first input light by three power splitters in two stages. A first pair of arms form an in-phase (I) branch in Mach-Zehnder modulation configuration that produces a first output with in-phase modulation components for the first input light into the TE Branch **2501**. A second pair of arms form a quadrature phase (Q) branch also in Mach-Zehnder modulation configuration plus a 90° phase shifter **2401** that generates a second output with quadrature phase modulation components for the first input light into the TE Branch **2501**. As the first output and the second output are combined, a TE-mode output light with four-level I/Q modulation of the first input light is generated in a TE-output line **2503**. Additionally, in the embodiment, a second I/Q modulator is configured, substantially the same as the first I/Q modulator, to have four linear waveguide arms with different phase delays to receive four equally split portions of the second input light into the TM* Branch **2502** by three power splitters in two stages. Similarly, a 90° phase shifter **2402** is used to combine the In-Phase and Quadrature components (I and Q) to generate a TM*-mode output light with four-level I/Q modulation of the second input light in a TM*-output line **2504**. Note, this TM*-mode output light still a TE-mode light originated from the tunable laser **1000**, even though it is labeled as TM*.

(45) Since the output light in both TE-output line and TM*-output line is TE mode, a polarization beam rotator combiner (PBRC) to combine them at the output to form a coherent optical signal. Referring to FIG. 1, the coherent transmitter block **2000** also includes a PBRC **2200** to rotate the TE-mode light in the TM*-output line **2504** to a TM-mode light combining with the TE-mode light in the TE-output line **2503**. Optionally, a TE attenuator **2301** and a TM* attenuator **2302** are two variable optical attenuators respectively inserted into the TE-output line **2503** and the TM*-output line **2504** to tune the polarization-dependent power loss in the two branches before the PBRC **2200**. The output of the PBRC **2200** is a fully coherent optical signal with mixed TE/TM modes in I/Q modulation. The output is then fed to the polarization independent semiconductor optical amplifier (SOA) **2100** to boost optical power of the coherent optical signal being transmitted out of the coherent transceiver. Optionally, residual polarization dependent gain in the polarization independent SOA **2100** can also be tuned with the variable optical attenuators **2301** and **2302** in the TE and TM* paths. Optionally, the use of the polarization-independent SOA **2100** at the output of the coherent transmitter block **2000** allows for wide range of output power from the coherent optical transceiver no matter the polarization state of the optical signal. Optionally, the polarization-independent SOA **2100** is provided as a flip-chip mounted on the silicon photonics substrate **100**. The flip-chip mounting of the SOA **2100** is similar to flip-chip integrations of the drive chip **2600** and the TIA chip **3400** with the silicon photonics substrate **100**.

(46) Optionally, the PBRC **2200** is substantially a same type of silicon-photonics device of the PBSR **3100** worked with reversed optical paths. In particular, referring to FIG. 2A, two TE-mode light waves, e.g., a TE-mode light with I/Q modulation and a TM*-mode light with I/Q modulation, are respectively loaded to two input ports **1901** and **1902** of the PBRC (which are two output ports for a PBSR). The TE-mode light loaded into the cross port **1901** is kept at the TE-mode at an

output port **1600** of the PBRC (which is an input port for the PBSR), while the TM*-mode light (also a TE-mode light) loaded in a bar port **1902** is rotated by 90 degrees to become a true TM-mode polarization at the output port **1600**.

(47) FIG. **3** is a simplified diagram of a silicon photonics tunable laser device according to an embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, a silicon photonics tunable laser device **10** includes a laser diode flip-chip **400**, a tunable filter **200**, and integrated couplers **301** and **302**, all of them being integrated into a single silicon photonics substrate **100**. The silicon photonics substrate **100** includes a patterned region **110** pre-fabricated with one or more vertical stoppers **112** and **113**, a plurality of alignment features **114**, **115** substantially distributed along a first direction *z*, and a bond pad **111** laid substantially between the one or more vertical stoppers **112** and **113**. The patterned region **110**, optionally, is configured to be a flat surface region a step lower than a rest flat surface region **120** separated by an edge **101** substantially along a second direction *x*. The integrated coupler **301** is disposed next to the edge **101** to couple with the tunable filter **200** formed in the silicon photonics substrate **100** so that the wavelength tuning can be achieved through the tunable filter **200**. The laser diode flip-chip **400** is configured to be attached into the patterned region **110** to bond with the bond pad **111** while rests against on the vertical stopper **112** and **113**. At the same time, the step section along the edge **101** also acts as an edge stopper against an end of the laser diode chip **400**. Eventually, laser light that has been tuned by the tunable filter **200** is outputted from an opposite facet of the laser diode chip **400**. Referring to via FIG. **1** and FIG. **3**, the integrated coupler **302** can be used to couple the laser output into a silicon wire waveguide in the same silicon photonics substrate **100** to deliver the laser light to either the receiver block **3000** and the transmitter block **2000**.

(48) In a specific embodiment, the tunable filter **200** is configured as a Vernier ring reflector filter. Optionally, the tunable filter **200** is a Si wire waveguide **220** fabricated in the silicon photonics substrate **100**. Optionally, the tunable filter **200** is formed in the rest flat surface region **120** beyond the patterned region **110**. The Si wire waveguide **220** includes at least two ring resonators **221** and **222**. Optionally, the two ring resonators are made with slightly different radii. Optionally, a first ring **221** is coupled to a reflector ring **223** which is also made by a linear section of the Si wire waveguide coupled to a ring-like structure via a 1-to-2 splitter. Optionally, a second ring **222** is coupled to the integrated coupler **300** via a linear wire waveguide **210** made by different material. Optionally, the linear wire waveguide **210** is a SiN based waveguide formed in the same silicon photonics substrate **100**.

(49) In the embodiment, the tunable filter **200** further includes a first ring heater (Ring1_HTR) **201** having a thin-film resistive layer overlying the first ring **221**, a second ring heater (Ring2_HTR) **222** having a resistive thin-film overlying the second ring **222**, and a phase heater (Phase_HTR) **203** overlying the reflector ring **223**. By changing temperature through changing voltages supplied to the two ring heaters (**201** and **202**), multiple resonate peak positions in transmission spectrum through each of the two ring resonators (**221** and **222**) can be tuned. Since the two rings have different radii, there is an offset between the two transmission spectra when they are superimposed (see FIG. **8**). As they go through the reflector ring **223**, a reflection light spectrum with a strong peak is produced (see FIG. **9**) and tunable by changing temperature through changing voltage supplied to the phase heater (**203**). Optionally, each heater is made by a resistive thin film geometrically covering each ring-shaped wire waveguide and terminated with two bond pads for bonding to an external power supply.

(50) In the embodiment, the laser diode chip **400** includes a gain region. The gain region includes an InP-based active region that is driven to produce a laser light. The laser light, as it is initially generated from the InP active region, is inputted via the integrated coupler **301** and the linear wire waveguide **210** into the tunable filter **200**. The light will pass through the at least two ring

resonators **222** and **221** and reflected by the reflector **223** back to the gain region of the laser diode chip **400**. The reflectivity spectrum, as shown in FIG. **10**, yields a strong peak of a laser light at a wavelength. The wavelength is tunable in a wide band range at least from 1560 nm to 1530 nm shown in FIG. **10** by tuning the phase heater **203**. The light is outputted as a laser light at a fixed wavelength when the round-trip cavity lasing condition is met if an integral of a whole light path of the tunable laser device **10** equals to $N2\pi$ (N is an integer).

(51) FIG. **4** is a schematic diagram showing a perspective view a laser diode chip flipped bonding to a silicon photonics substrate according to an embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. Referring to FIG. **4**, a section of a substrate **100** is shown to include a patterned region **110** out of rest flat surface region **120**. The patterned region **110** is configured to be a flat region being a step lower than rest flat surface region **120**. Along one step an edge stopper **101** is formed facing a first direction z . Optionally, the step is along a second direction x perpendicular to the first direction z . On the patterned region **110**, there is a bond pad **111** formed along the first direction z . On both sides of the bond pad **111**, there are two vertical stoppers **112** and **113** which are two thin plates having certain heights. Two extended parts **117** of the bond pad **111** are used for bonding with an external source. Further along these vertical stoppers, a plurality of alignment features **115** are formed. Optionally, an alignment feature **115** includes multiple fiducials lined in one or two rows along the first direction z .

(52) In the embodiment, a laser diode chip **400** can be pre-fabricated with a gain region **410** and a metallic electrode **411** formed on top in an elongated shape. The gain region **410** is formed from one edge to another of the chip. Optionally, the metallic electrode **411** is formed to be in contact with a P-side layer of an active region made by InP-based P-N junction quantum well structure. On two sides of the metallic electrode **411**, an alignment feature **415** is formed on the laser diode chip and configured to match the plurality of fiducials of the alignment feature **115** on the patterned region **110**. Referring to FIG. **4**, the laser diode chip **400** is a flip chip bonded onto the patterned region **110** of the substrate **100**. The configurations for both the patterned region **110** and the laser diode chip **400** allow the latter is rest against the vertical stoppers **112** and **113** of the former with one edge of the latter against the edge stopper **101** of the former as the alignment feature **415** of the latter is engaged with the plurality of fiducials **115** of the former.

(53) FIG. **5** is an exemplary diagram of a wavelength tuning map of a silicon photonics tunable laser according to an embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, the diagram is plotted as spectrum frequency 2D map varied with powers supplied to two power supplies for the two ring heaters in the tunable filter mentioned above. The dashed rectangle gives a possible tuning range. As the two ring resonators are provided with different radii, it effectively yields an extended tunable range for the spectrum wavelength as the two transmission spectra are superimposed when the two ring resonators are physically coupled, for example, in a Vernier ring filter configuration (see FIG. **8**). Optionally, the long side of the rectangle provides a relative rough tuning range of wavelength for more than 50 nm, e.g., from ~1520 nm to ~1575 nm, and the short side of the rectangle provides a relative finer tuning range of wavelength for ~10 nm, e.g., from ~1565 nm to ~1575 nm.

(54) FIG. **6** is a simplified diagram of a silicon photonics tunable laser device according to another embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, the tunable laser device **40** includes a first laser diode chip **430** bonded onto a first patterned region **410** of a substrate **100**, a second laser diode chip **440** bonded onto a second patterned region **420** of the substrate **100**, a tunable filter **470** coupled to a

first gain region **435** of the first laser diode chip **430** via a first integrated coupler **451** and coupled to the second gain region **445** of the second laser diode chip **440** via a second integrated coupler **460**, a wavelength locker **480** configured to lock the wavelength of a reflected light from the second gain region **445** through the tunable filter **470**, and a laser output is realized at an end facet **432** of the first gain region **435** where a third integrated coupler **452** can be employed for coupling the laser output to a silicon wire waveguide. Referring to FIG. **1** and FIG. **6**, the laser output from the tunable laser **1000** can be coupled via the integrated coupler **452** to a waveguide in the silicon photonics substrate **100** which further delivers to both the coherent receiver block **3000** and the coherent transmitter block **2000** via waveguide through a power splitter **1001**.

(55) Referring to FIG. **3** and FIG. **6**, each of the first laser diode chip **430** and the second laser diode chip **440** of the tunable laser **40** is substantially configured to be same as one laser diode chip **400** that has a metallic electrode in contact with a P-type layer of corresponding active region and is flipped down to have the metallic electrode bonded with a bond pad of the first patterned region **410** and the second patterned region **420** in the same substrate **100**. In particular, the first laser diode chip **430** is configured to be a cavity having a first end facet **431** and a second end facet **432** with the gain region **435** as a wire waveguide along an active region bounded between them. As the first laser diode is flipping (P-side down) bonded to the first patterned region **410**, the first end facet **431** is against an edge stopper associated with the first patterned region **410** to be aligned with the first integrated coupler **451**. The silicon wire waveguide associated with the first gain region **435** is configured to be in a curved shape with a non-perpendicular angle relative to each of the first end facet **431** and the second end facet **432** to reduce direct back reflection of the light by the corresponding end facet. Optionally, the first end facet **431** includes an anti-reflective coating. Optionally, the second end facet **432** is also coated with an anti-reflective coating for facilitating laser output. The first laser diode chip **430** is driven by a driver to generate a light as an input light into the tunable filter **470** via first waveguide **491**. The second laser diode chip **440** is configured to be a cavity having a third end facet **441** and a fourth end facet **442** with the second gain region as a silicon wire waveguide along an active region bounded between them. As the second laser diode chip is flipping (P-side down) bonded to the second patterned region **420**, the first end facet **441** is against an edge stopper associated with the second patterned region **420** to be aligned with the second integrated coupler **460**. Optionally, the third end facet **441** is coated by an anti-reflective coating and the fourth end facet **442** is coated by high-reflective coating to enhance reflection of the input light. The reflected light can pass through the second integrated coupler **460** back to the tunable filter **470** via a second waveguide **492** so that the cavity of the second gain region **445**, in addition to provide gain to the reflected light, acts also as a ring reflector of the tunable filter in substantially a Vernier ring reflector configuration.

(56) In the embodiment, the tunable filter **470** of the tunable laser **40** is a Si wire waveguide fabricated in the substrate **100** particularly in a region **120** beyond the first patterned region **410** and the second patterned region **420**. The tunable filter **470** includes two ring resonators, a first ring resonator **471** and a second ring resonator **472** coupled to each other. The two ring resonators are coupled to the first waveguide **491** via a 2-to-1 coupler **473** and coupled to the second waveguide **492** via another 2-to-1 coupler **474**. Optionally, the 2-to-1 coupler is still in waveguide form with one port in one end and two ports in opposite end. Optionally, it is a splitter from one-port end to two-port end or a combiner from two-port end to the one-port end. Both the first waveguide **491** and the second waveguide **492** are fabricated in the region **120** of the substrate **100** to couple respective with the first integrated coupler **451** and the second integrated coupler **460**. The first integrated coupler **451** is disposed next to the edge stopper associated with the first patterned region **410**. The second integrated coupler **460** is disposed next to the edge stopper associated with the second patterned region **420**. Optionally, each of the first waveguide **491** and the second waveguide **492** is made by SiN material embedded in a Si-based substrate **100**. Optionally, the wire waveguide of the tunable filter **470** is made by Si material.

(57) Referring to FIG. 4, the tunable filter **470** of the tunable laser **40** further includes a first heater (Ring1_HTR) having a resistive thin-film overlying the first ring resonator **471**, a second heater (Ring2_HTR) having a resistive thin-film overlying the second ring resonator **472**, a third heater (Phase_HTR) having a resistive thin-film overlying a phase shifter section **475** of the Si wire waveguide that is connected with the second waveguide **492**. These heaters are configured to change temperature to cause a change of transmission spectrum of the light passing through respective ring resonators. Each transmission spectrum of the ring resonator has multiple resonate peaks (see FIG. 6). In an embodiment, the two ring resonators, **471** and **472**, are provided with slightly different radii, then an offset between the two transmission spectra exists when they are superimposed (see FIG. 6). The first heater and the second heater can be controllably change temperatures of the respective first ring resonator and the second ring resonator to cause respective resonate peaks to shift to provide an extended tunable range of the wavelengths of those resonate peaks. After the input light passing through the tunable filter **470** is reflected back by the cavity of the second gain region **445**, a reflectivity spectrum gives a stronger central peak (see FIG. 9), which is further tunable by changing temperature of the phase shifter section **475** using the third heater. In this case, the second gain region **445** acts as a ring reflector as the tunable filter **470** is configured to be a Vernier ring reflector while the phase shifter section **475** is formed next to the reflector instead of separated from the reflector as shown in FIG. 10.

(58) Optionally, the tunable filter **470** of the tunable laser **40** includes a fourth heater having a resistive thin-film overlying a section of one branch of the 1-to-2 coupler **473** for finely balance power of input light split from the first waveguide **491** into the two branches respectively coupled to two ring resonators **471** and **472**.

(59) In the embodiment, the wavelength locker **480** is configured to be a delay line interferometer (DLI) based on Si waveguide formed in the substrate **100**. Optionally, the wavelength locker **480** includes an input port coupled via a splitter to the first waveguide **491** to receive the reflected light from the tunable filter **470**. Optionally, one end of the input port is a SiN waveguide coupled to the first waveguide **491** which is also made by SiN material. Another end of the input port connects to a 1-to-2 splitter **481** to guide one part of the light into a monitor port PM0 and another part of the light into the DLI via another 1-to-2 splitter **482**. The light then comes out of the DLI via a 2-to-2 splitter **483** to a first interference output port PM1 and a second interference output port PM2. In the embodiment, the wavelength locker **480** is pre-calibrated to set the DLI for locking the wavelength (of the reflected light from the tunable filter) to certain channel wavelengths of a wide band. Optionally, the channel wavelengths supplied by the tunable laser **40** are ITU channels in C-band for DWDM application. Of course, the disclosure of the tunable laser **40** can be applied to O-band light source for CWDM application. Optionally, the tunable laser **40** is integrated in a coherent transceiver a coherent receiver block and a coherent transmitter block substantially on a same silicon photonics substrate (see FIG. 1). Optionally, each of the monitor port PM0, the first interference output port PM1, and the second interference output port PM2 is terminated with a photodiode for measuring light power in terms of photocurrent. A differential signal characterized by a photocurrent difference between PM1 and PM2 over the sum of them is collected to be an error signal fed back to drivers for the first laser diode chip and the second laser diode chip to adjust wavelength of light. Ideally, when the wavelength is adjusted or locked to a desired ITU channel pre-calibrated for the wavelength locker, the error signal should be zero, i.e., PM1=PM2.

(60) In another aspect, the present disclosure also provides a method for tuning wavelengths of a silicon photonics based tunable laser device described hereabove. FIG. 7 is a flowchart of a method for tuning wavelength of a laser output of the silicon photonics tunable laser device according to an embodiment of the present invention. As shown, the method includes a step of generating a light with a wavelength near ITU channels (for example, in C-band) in a dual-gain configuration including a first active region and a second active region. Referring back to FIG. 6, in a specific embodiment, a first laser diode chip having a first metallic electrode in contact with a P-type layer

of the first active region is provided and a second laser diode chip having a second metallic electrode in contact with a P-type layer of the second active region is provided. Further, the first laser diode chip to have the first metallic electrode is flipping bonded with a bond pad in a first patterned region of a substrate to align the first active region to a first integrated coupler and the second laser diode chip to have the second metallic electrode is flipping bonded with a bond pad in a second patterned region of the substrate to align the second active region to a second integrated coupler. The first active region and the second active region is connected via the silicon photonics tunable filter to form a combined resonate cavity. Referring to FIG. 7, the method includes driving the first laser chip and the second laser diode chip to generate the light in the combined resonate cavity. In particular, the method includes a step of inputting the light with gain from a first active region of a first laser diode chip driven by its driver. The input light with gain from the first active region is coupled into a first integrated coupler into the first waveguide which guides the input light to the silicon photonics based tunable filter. Additionally, the input light from the first active region passes through the first integrated coupler via a first waveguide into the tunable filter and it further passes via a second waveguide and through the second integrated coupler into the second active region and reflected with additional gain therefrom, achieving dual gains through the round-trip path of the combined resonate cavity.

(61) Referring to FIG. 7, the method further includes reflecting the light with additional gain from a second active region of a second laser diode chip. The reflected light with additional gain from the second active region further passes through the second integrated coupler and back to the tunable filter via a second waveguide.

(62) Additionally, the tunable filter is configured to have a first ring resonator Ring1, a second ring resonator Ring2, and a Phase shifter section. The method further includes a step of setting respectively a first heater associated with the first ring resonator, a second heater associated with the second ring resonator, and a third heater associated with the phase shifter to set the wavelength near an ITU channel. Respectively, the first heater, the second heater, and the third heater are configured to be resistive thin-films formed in the substrate to cover at least partially the first ring resonator Ring1, the second ring resonator Ring2, and the Phase shifter section. Each of these heaters can be controlled by voltages applied to two coupling electrodes from an external power supply. In a specific embodiment, the step includes reading voltages respectively set from the first heater, the second heater, and the third heater from a preset look-up-table (LUT). The voltages read from the LUT are substantially correlated with the corresponding ITU channel. For example, some specific voltage values are preset for wavelength at 1535 nm in C-band. Further, the step includes applying voltages read from the LUT respectively to the first heater and the second heater to respectively set two transmission spectra of the first ring resonator and the second ring resonator to obtain a synthesized spectrum with a strong peak wavelength in an extended tunable range. For example, the extended tunable range can be varied from 1520 nm to 1570 nm. In another example, when the gain profile is relatively limited, the extended tunable range can be at least varied from 1535 nm to 1565 nm. Furthermore, the step includes applying a voltage read from the LUT to the third heater to set a phase of a reflectivity spectrum with the strong peak wavelength in the extended tunable range. The reflection spectrum is set substantially based on the synthesized spectrum.

(63) Referring to FIG. 7, the method further includes monitoring photocurrents at a monitor port split from the input port, a first interference output port, and a second interference output port of the wavelength locker (see FIG. 6) based on the light reflected from the second active region and filtered by the tunable filter. Each of the monitor port, the first interference output port, and the second interference output port is respectively terminated with a photodiode (such as PM0, PM1, and PM2, see FIG. 6). Each of these photodiodes generates a photocurrent as a measurement of the light power thereof, which can be monitored in real time. An error signal based on a differential light power between the first interference output port and the second interference output port can be

used as a feedback for tuning light wavelength to be locked to a pre-calibrated wavelength, e.g., an ITU channel.

(64) Furthermore, the method includes tuning the first heater and the second heater to coarsely tune the transmission spectrum through each of the first ring resonator and the second ring resonator until the photocurrents at the first interference output port and the second interference output port are equal. Since slightly different radii are assigned for the first ring resonator and the second resonator, an offset exists between the two transmission spectra. A synthesized spectrum can be obtained by superimposed the two transmission spectra, which includes at least one strong peak as two transmission peaks of two ring resonators are falling to a same wavelength. By tuning both the first heater and the second heater, the position of this strong peak in the synthesized spectrum is shifted in the extended tunable range. When the photocurrents at the first interference output port and the second interference output port are equal, it means the peak wavelength is substantially tuned to match a pre-calibrated wavelength locked by the wavelength locker as the error signal becomes zero. Of course, the wavelength locker **480** can be configured to in different manners for achieving the wavelength locking function in combination with the tunable filter **470** configured as a Vernier ring reflector. Many of the different silicon-photonics-based wavelength locker configurations can be referred to a U.S. Pat. No. 10,056,733 commonly assigned to Inphi Corporation.

(65) The method further includes tuning the third heater to finely tune reflection spectrum by maximizing the photocurrent of the monitor port of the wavelength locker representing a maximum gain from a round trip cavity lasing condition associated with both the first active region and the second active region. The Phase shifter section of the tunable filter is located at a straight section of Si wire waveguide outside the two ring resonators. When the third heater, which is placed at least partially over the Phase shifter section, is tuned to change temperature of the Phase shift section, a phase of the reflected light can be tuned in accordance with the whole round-trip path of the light between the first active region and the second active region. The lasing condition is a maximum gain obtained under the physical setup of the optical path between the first active region and the second active region and phase optimized by the Phase shift section, which is characterized by the maximum power measured by photocurrent at the monitor port of the wavelength locker. The enlarged cavity with two active regions over a single active region certainly enhances lasing power of the tunable laser device.

(66) FIG. **8** is an exemplary diagram of two superimposed transmission spectra of respective two ring resonators with different radii of a tunable filter in the silicon photonics tunable laser device according to an embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, each transmission spectrum of a ring resonator contains multiple peaks with a certain spacing depended on the radius of the ring. Since, two different radii are provided for Ring1 and Ring2, which results in two different spectra-free-ranges (SFRs), there is an offset between the two transmission spectra. Yet, two specific peaks respectively from the two transmission spectra may fall to a substantially common wavelength, for example, ~1540 nm.

(67) FIG. **9** is an exemplary diagram of a reflectivity spectrum of a reflector coupled to the two ring resonators of the tunable filter according to the embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. After the two transmission spectra are superimposed as light passing the two ring resonators is recombined in the single straight section of the wire waveguide and reflected from the second active region, a reflection spectrum is obtained which is substantially based on a synthesized spectrum of the two transmission spectra. As shown, the reflection spectrum is characterized by at least one strong peak at a wavelength, e.g., ~1540 nm.

(68) FIG. 10 a simplified block diagram of the tunable filter including two ring resonators, a reflector, plus a phase shifter in a Vernier ring reflector configuration according to an embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, the tunable filter includes three sections: two ring resonators and phase control sections. The two ring resonators have slightly different FSRs, which allows for extension of the tuning range to the lowest common multiple of the FSRs through the Vernier effect. One phase control section is a loop reflector formed by a loop waveguide and a directional coupler. Optionally, the loop reflector can be replaced by a cavity facet of another laser diode chip. Another phase control section can be simply a section of waveguide with an added heater for directly tuning phase based on thermal optical effect. A reflection spectrum is substantially a synthesized spectrum obtained by superimposing two ring spectra. External laser cavity is configured between the facet of the (first laser diode chip) active region and a loop reflector. When the peaks of the transmission spectrum through the rings are identical and the phase is adjusted on the peak of the rings, lasing operation is occurred. Of course, there are multiple variations of the tunable filter configuration in terms of a setup of the ring resonators and phase control sections, resulting different synthesized spectrum.

(69) FIG. 11 is an exemplary diagram of two synthesized spectra respectively corresponding to wavelengths being tuned from 1555 nm to 1535 nm by tuning the tunable filter in Vernier ring reflector configuration and corresponding gain profile from 1530 nm to 1570 nm according to the embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, the synthesized spectrum is characterized by a strong peak due to superimposing two transmission spectra with common multiple peak wavelengths for different SFRs. The peak position, or wavelength value, can be tuned by tuning the Vernier ring reflector around an optimal central position in an extended tunable range.

(70) FIG. 12 is an exemplary diagram of laser spectra outputted by the silicon photonics tunable laser device with laser wavelength being tuned from 1555 nm to 1535 nm according to an embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, the laser wavelength is indicated by the peak of the spectra, which can become tunable across C-band (or other wide band) by tuning the Vernier ring reflector response. In the example, the laser wavelength is tuned from 1555 nm to 1535 nm. The tunable filter based on Vernier ring reflector configuration acts as wavelength selective filter for the gain profile. The center or optima position of the gain profile can be initially preset by setting the optimal temperatures using pre-calibrated voltages supplied to the resistive heaters associated with the Ring1, Ring2, and Phase shift section. For example, the pre-calibrated voltages can be stored in a look-up-table of a memory, which can be read every time for initialing the silicon photonics based tunable laser device. Coarse wavelength tuning can be achieved by changing temperatures around the Ring1 and the Ring2 to tune the wavelength in an extended tunable range around the optimal gain profile position set by the initial settings of the heaters associated with the Ring1, Ring2, and the Phase shift section. Fine wavelength tuning can be done by changing temperature around the Phase shift section. The laser regions also have a wavelength dependent gain profile which is much wider than those from the rings.

(71) FIG. 13 is a schematic diagram of three types of an integrated coupler based on SiN in Si waveguide according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. Like the FSRs, the coupling coefficients of the tunable filter with two ring resonators are important parameters in order to achieve large modal gain-difference for obtaining enough high side-mode suppression ratio to other

modes. One key factor to determine the coupling coefficient is the alignment of light mode confined in the integrated coupler with the incoming laser light beam. On the other hand, tolerance to misalignment existed with different integrated coupler designs would be an advantage for enhance productivity of the tunable laser device.

(72) Referring to FIG. 13, a first type of integrated coupler design is an Inverse Nanotaper structure made by SiN material embedded in a Si waveguide with a sharp needle pointing toward a waveguide end. A small mode diameter is shown for this design, giving large coupling loss in response to a relatively small misalignment. A second type of integrated coupler design is a Trident structure made by SiN material embedded in a Si waveguide. The SiN Trident structure includes a SiN nanotaper sandwiched in partial length laterally by two SiN symmetrical nanotapers that are extended up to the waveguide end of the coupler. A large mode diameter for this design is shown, giving lower coupling loss induced by misalignment. A third type of integrated coupler design is a Fork structure made by SiN material embedded in a Si waveguide. The Fork structure includes a SiN nanotaper sandwiched in full length laterally by two SiN linear stripes up to the waveguide end. It has a medium sized mode diameter yet giving a smallest coupling loss especially for smaller misalignment between the light mode and the laser spot.

(73) FIG. 14A is an exemplary diagram of a relationship between coupling loss and a lateral misalignment for an integrated coupler coupled between the tunable filter and a laser diode chip according to an embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, SiN Trident and SiN Fork type integrated couplers offset lower coupling loss alternative to SiN Inverse Nanotaper type integrated coupler with comparable tolerance to misalignment in lateral axis between a laser beam coming out of the laser diode chip and mode diameter of the integrated coupler. In the example, the coupling loss from mode mismatch based on $2.5\text{ }\mu\text{m}$ laser diode spot size. FIG. 14B is an exemplary diagram of a relationship between coupling loss and a vertical misalignment for the integrated coupler coupled between the tunable filter and a laser diode chip according to the embodiment of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, SiN Trident and SiN Fork type integrated couplers offset lower coupling loss alternative to SiN Inverse Nanotaper type integrated coupler with comparable tolerance to misalignment in vertical axis between a laser beam coming out of the laser diode chip and mode diameter of the integrated coupler. Additionally, the SiN Fork type integrated coupler offers more compact design than SiN Trident type integrated coupler with lower coupling loss (e.g., less than 1 dB). In an example, the Fork type integrated coupler has just half length of the Trident type integrated coupler. In an example, the coupling loss for the Fork type is just half of that for Trident type when the misalignment is less than $0.6\text{ }\mu\text{m}$ in both lateral or vertical direction.

(74) In an embodiment, the coherent optical transceiver is integrated in a single chip. FIG. 15A shows a perspective view of a coherent optical transceiver chip is integrated on a silicon photonics substrate. FIG. 15B shows a side view of the coherent optical transceiver chip. Referring to FIG. 15A, a silicon photonics substrate **100** is provided its top surface as a substrate for integrating all components of a coherent optical transceiver chip **5000**. The coherent optical transceiver chip **5000** has a fiber-coupler **520** configured to hold a first fiber **511** and a second fiber **512** respectively coupled with the silicon photonics substrate **100**. The coherent optical transceiver chip **5000** includes a tunable laser device **1000** including two laser diode (LD) chips, a first LD **430** and a second LD **440**, both flip-mounted onto a first region of the silicon photonics substrate **100**. Both LD chips **430** and **440** are coupled with a wavelength tunable section embedded in the first region as depicted in the FIG. 15A. The tunable laser device **1000** is substantially similar to the tunable laser **40** shown in FIG. 6. The two laser diode chips **430** and **440** are configured to have p-side electrodes of respective two gain regions (referred to **430** and **440** too) facing down to mount on

the silicon photonics substrate **100**. The substrate **100** is prepatterned with a plurality of surface fiducials, vertical stoppers, edge stoppers, and bond pads for mounting and aligning the LD chips. The mounting of the LD chips **430** and **440** allows respective backend facets of the two gain regions to be aligned optically with the wavelength tuning filter **470** and a wavelength locker **480**. At the same time, the mounting of the LD chip **430** allows its front-end facet of the first gain region to output the laser light coupled into a waveguide that leads to a silicon photonics circuit. Optionally, the wavelength tunable filter **470** and the wavelength locker **480** are directly made by wire waveguides in the first region of the silicon photonics substrate **100**. Optionally, they are made by silicon and silicon related materials embedded into the silicon photonics substrate **100**. In particular, the wavelength locker **480** includes a delay-line interferometer and multiple optical power monitors made by silicon or silicon nitride waveguides. The wavelength tunable filter **470** includes two or more micro-ring resonator waveguides **471** and **472** followed by linear waveguides **491** and **475** respectively coupled to the first gain region **430** and the second gain region **440**. The two or more ring resonators have slightly different radii to allow a light coupled from the two gain regions being reflected therein multiple times with at least 90% reflectivity and being tuned in an extended wavelength range of synthesized spectrum by at least one resistive heater partially overlying each of the two or more micro-ring resonator waveguides and a heater overlying a section of linear waveguide **475** as a phase shifter. Optionally, the power supplies for the resistive heaters and LD driver are provided through electrical connections embedded in the substrate **100** and multiple through-silicon vias (TSVs) in the substrate **100** to connect with some of conducting bumps of the board grid array (BGA) at a bottom surface of the substrate **100** (as seen in FIG. **15B**).

(75) In the embodiment, the coherent optical transceiver chip **5000** further includes a transimpedance amplifier (TIA) chip **3400** flip-mounted on a second region of the silicon photonics substrate **100**. The TIA chip **3400** is electrically coupled to corresponding electrical connections embedded in the substrate **100** or through some TSVs connected to BGA at the bottom surface of the substrate **100**. A printed circuit board (PCB) can be used to mount the silicon photonics substrate **100** to provide desired electrical connection for the TIA chip for operating the optical transceiver **5000**. Optionally, the TIA chip **3400** is configured to prepare some voltage signals converted from some hybrid current signals detected from incoming coherent light signals. Additionally, the coherent optical transceiver chip **5000** includes a driver chip **2600** flip-mounted on a third region of the silicon photonics substrate **100**. The driver chip **2600** is electrically coupled to corresponding electrical connections embedded in the substrate **100** or through some TSVs connected to BGA at the bottom surface of the substrate **100**, and to utilize the PCB for completing the electrical connections.

(76) Furthermore, the coherent optical transceiver chip **5000** includes a silicon photonics circuit integrated with the tunable laser device **1000**, the TIA chip **3400** and the driver chip **2600**. The silicon photonics circuit is directly formed in a fourth region of the silicon photonics substrate **100** which is substantially sunken into the substrate **100**. Optionally, the silicon photonics circuit includes several silicon photonics devices respectively formed in different geometrical portions of the fourth region. Functionally, as shown in FIG. **1**, the silicon photonics circuit includes mainly a receiver block **3000** and a transmitter block **2000**. Referring to FIG. **1** and FIG. **15A**, the receiver block **2000** is configured to have a waveguide (not explicitly shown) coupled with the first fiber **511** to receive a coherent input light signal and to have another waveguide (not explicitly shown) coupled with the tunable laser device **1000** to receive a first portion of a laser light as local oscillator signals to assist detections of both TM-mode and TE-mode light signals in the coherent input light signal by the TIA chip **3400**. The transmitter block **2000** is configured to use the driver chip **2600** to drive modulations of a second portion of the laser light from the tunable laser device **1000** to generate a coherent output light signal outputted to the second fiber **512**.

(77) Referring to FIG. **15A**, although no details are explicitly shown, the silicon photonics circuit

includes a first power splitter **1001** formed in the substrate **100** having an input waveguide **1010** coupled and aligned to the front-end facet of the first gain region **430** to receive the laser light outputted therefrom. The first power splitter **1001** is configured to split the laser light to two portions, a first portion outputted to a first output waveguide **1011** and a second output waveguide **1012** with an X:Y ratio ranging from 10:90 to 50:50. The silicon photonics circuit includes a sub-circuit having at least a polarization beam splitter rotator (PBSR), a second power splitter, and a pair of optical receivers, configured as a receiver block **3000** shown in FIG. 1, integrated in one sub-region **3101** of the fourth region in the silicon photonics substrate **100**. Functionally, the polarization beam splitter rotator **3100** has an input waveguide directly coupled to the first fiber **511** at an edge of the silicon photonics substrate **100** to receive a coherent input light signal. The PBSR **3100** is configured as a shaped silicon waveguide (see FIG. 2A) to receive the coherent input light signal and output the TE-mode light signal of the coherent input light signal and a TM*-mode light signal to two separate optical paths respectively fed to the two 90° hybrid optical receivers **3200** and **3300**. The TM*-mode light signal is essentially a TE-mode light signal rotated 90° from the TM-mode light signal of the coherent input light signal. The second power splitter **3001** is essentially a 3 dB coupler to split the first portion of the laser light coming from the first output waveguide **1011** of the first power splitter **1001** to two equal halves. Each half of the laser light is used as a local oscillator signal inputted to a respective one of the two 90° hybrid optical receivers **3200** and **3300** to combine with the respective TE- and TM*-mode light signal to form a first hybrid light signal and a second hybrid light signal. Correspondingly, the two 90° hybrid optical receivers **3200** and **3300** includes photo diodes configured to respectively convert the first hybrid light signal and a second hybrid light signal to a first hybrid current signal and a second hybrid current signal.

(78) Referring to FIG. 15A, the silicon photonics circuit also includes a polarization beam rotator combiner (PBRC) and a pair of optical modulators configured as a transmitter block **2000** shown in FIG. 1, integrated respective two sub-regions **2301** and **2302** of the fourth region in the silicon photonics substrate **100**. Particularly, each of the pair of optical modulators includes a silicon-waveguide based Mach-Zehnder interferometer configured as in-phase/quadrature-phase modulator having one in-phase branch and one quadrature-phase branch with a 90° phase shifter, each branch being biased by voltages provided from the driver chip **2600** to drive modulations to light received from the second portion of the laser light coming from the second output waveguide **1012** of the first power splitter **1001** and passing through the silicon waveguides of the modulator itself. Each modulator is configured to output a modulated light signal with I/Q four-level modulations substantially in TE-mode polarization originated from the laser light from the tunable laser device **1000**. The PBRC **2200** is coupled to the two in-phase/quadrature-phase modulators to receive the two TE-mode modulated light signals, output one retaining the TE-mode modulated light signal while another one being rotated to a TM-mode modulated light signal, and combine the TE-mode modulated light signal with the TM-mode modulated light signal to a coherent modulated light signal. Referring to FIG. 15A, the silicon photonics circuit further includes a polarization-independent semiconductor optical amplifier (PI-SOA) **2100** as a chip flip-mounted on the silicon photonics substrate **100** near an edge to couple with the PBRC **2200** at one end and couple with the second fiber **512** at an opposite end. The PI-SOA **2100** is configured to provide a wide range of output power of the coherent modulated light signal. Further, the PI-SOA **2100** delivers the amplified coherent light signal to the second fiber outputted to the output port as a coherent output light signal.

(79) FIG. 15B shows a side view of the coherent optical transceiver chip **5000** of FIG. 15A. As shown, the coherent optical transceiver **5000** is integrated on the single silicon photonics substrate **100**. The silicon photonics substrate **100** is a silicon photonics chip comprising waveguides made by silicon-related materials formed in a silicon-on-insulator substrate. A top surface of the silicon photonics substrate **100** is configured to allow multiple functional chips like TIA chip **3400**, driver

chip **2600**, laser diodes chip **430** and **440**, and semiconductor optical amplifier **2100** to be mounted at respective regions. Optionally, each of these chips is flip-mounted with conductive bumps **170** facing down to bond with pre-formed bumps on the respective regions. Optionally, a plurality of conductive through-silicon vias can be formed through the silicon photonics substrate **100** and filled with conductive materials for connecting those chips mounted on top thereof to some of board grid array **160** at bottom thereof. Optionally, the board grid array **160** is designed for mounting this integrated coherent optical transceiver **5000** on the silicon photonics substrate **100** onto a printed circuit board in a modular package.

(80) In another aspect, the present disclosure provides a compact package for an integrated coherent optical transceiver formed on a single silicon photonics chip. FIG. **16A** is a schematic diagram of an open package of an integrated coherent optical transceiver according to an embodiment of the present invention. FIG. **16B** shows a closed package of the integrated coherent optical transceiver of FIG. **16A** according to the embodiment of the present disclosure. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown, the closed package **6000A** is a fully cased coherent optical transceiver. The open package **6000** is with a lid member **150** being removed. The package **6000** includes an integrated coherent optical transceiver **5000** mounted on a printed circuit board (PCB) **600** in a metal case. The metal case is, in addition to the lid member **150**, assembled together with a pair of side members **140** naturally connected by a joint member **141** near a front-end region of the case and a bottom member **130** extending the joint member **141** to a back end **131** of the case. The pair of side members **140** has a pair of clip structures **145** to couple with the lid member **150**. The PCB **600** has its board body supported on the bottom member **130** with multiple conductive pins **610** formed at an end region of the board body substantially located at the back-end region **131** of the case. The package includes a first optical connector receptor **501** configured as an input port and a second optical connector receptor **502** configured as an output port for the transceiver **6000**. Both input port and the output port are located on the joint member **141** before the front end of the PCB board body **600**.

(81) The integrated coherent optical transceiver on the silicon photonics substrate **100** is substantially the coherent optical transceiver chip **5000** shown in FIG. **15A**. Details of the coherent optical transceiver chip **5000** can be found in several earlier paragraphs. The first optical connector receptor **501** has its back end coupled with a first fiber **511** and the second optical connector receptor **502** has its back end coupled with a second fiber **512**, both being held by a fiber-coupler **520** to respectively couple and optically align with an input waveguide into a silicon photonics circuit built in the silicon photonics substrate **100** and an output waveguide connected from a polarization-independent semiconductor optical amplifier (PI-SOA) **2100**. Optionally, the first optical connector receptor **501** has its front end configured to mate with an optical connector connected to an input fiber delivering a coherent input light signal with mixed TE- and TM-mode from an optical communication system. The second optical connector receptor **502** has its front end configured to mate with an optical connector connected to an output fiber for outputting a coherent output light signal with four-level I/Q modulation in both TE-mode and TM-mode polarizations. Depending on applications, the integrated coherent transceiver package **6000A** can be configured with a compact form factor that adapts with any system design for coherent optical communication.

(82) In an alternative aspect, the present disclosure provides a light engine device including an optical coherent transceiver integrated on a semiconductor substrate member. For example, the optical coherent transceiver is provided as the integrated coherent optical transceiver shown in FIG. **1**. For example, the substrate member is provided as one like the substrate **100** of FIG. **15B**. The light engine device includes a substrate member comprising a surface region. The light engine device further includes an optical input configured to an incoming fiber device and an optical output configured to an outgoing fiber device. Additionally, the light engine device includes a

transmit path provided on the surface region. The transmit path includes a polarization independent optical amplifier device coupled to the optical output. The transmit path also includes a polarization beam rotator combiner device coupled to the polarization independent optical amplifier and coupled to the optical output. The transmit path further includes a dual polarization I/Q Mach Zehnder modulator device coupled to the polarization beam rotator combiner device and coupled to the optical output. The transmit path again includes a driver device coupled to the dual polarization I/Q Mach Zehnder modulator device and configured to drive an electrical signal to the dual polarization I/Q Mach Zehnder modulator. Furthermore, the transmit path includes a tunable laser device comprising a laser diode chip having a gain region with a p-side electrode flipped down and mounted on the substrate member. The gain region is coupled with a wavelength tuning section formed in the substrate member to tune wavelengths of a laser light outputted from the gain region to a waveguide in the substrate member. Moreover, the transmit path includes a first power splitter coupled to the waveguide to split the laser light to a first light and a second light. The second light is coupled to the dual polarization I/Q Mach Zehnder modulator device. Further, the light engine device also includes a receive path provided on the surface region. The receive path includes a second power splitter coupled to the first light. The receive path further includes a pair of 90° hybrid receivers. Each of the pair of 90° hybrid receivers includes a photo detector device and a hybrid mixer device, coupled respectively to two outputs of a polarization beam splitter rotator in the substrate member to receive the optical input and to two outputs of the second power splitter to receive the first light from the tunable laser device for assisting detections of a transverse electric (TE) mode signal and a transverse magnetic (TM) mode signal in the coherent input signal. Furthermore, the receive path includes a transimpedance amplifier coupled to each of the 90° hybrid receivers and coupled to each of the photo detector devices that convert a combination of the first light with the optical input into an electrical signal to be transmitted to using the transimpedance amplifier device. The light engine device further includes a heterogeneous integration configured using the substrate member, the transmit path, and the receive path to form a single silicon photonics device.

(83) Optionally, the substrate member comprises a silicon substrate. Optionally, the transimpedance amplifier is made of a silicon germanium bipolar technology. Optionally, the transimpedance amplifier is made of a silicon CMOS technology. Optionally, the transimpedance amplifier is made of an indium phosphide technology. Optionally, the transimpedance amplifier is made of a gallium arsenide containing technology. Optionally, the data center is configured for a social networking platform, an electronic commerce platform, an artificial intelligence platform, or a human tracking platform.

(84) Optionally, the light engine device is coupled to a major substrate member. The major substrate member, for example, provided as a circuit board **600** of FIG. **16A**, is configured to mount the light engine device, a digital signal processing device coupled to the light engine device, a power supply coupled to the light engine device and the digital signal processing device, a micro-controller device coupled to the light engine device to provide one or more controls using one or more control signals to the light engine device, an electrical input and output configured to the light engine device, the digital signal processing device, the power supply, and the micro-controller device; and a mechanical and electrical configuration including the light engine device, the digital signal processing device, the power supply, the micro-controller device, and the electrical input and output configured to the light engine device, the digital signal processor device, the power supply and the micro-controller device.

(85) Optionally, the light engine device and the major substrate member are configured as a pluggable device. Optionally, the light engine device and the major substrate member together are configured on a system board member of a communication system intended for data transmitting and receiving. Optionally, the system board member is provided in a switch system apparatus, the switch system apparatus being spatially disposed in a data center. Optionally, the data center is

configured for a social networking platform, an electronic commerce platform, an artificial intelligence platform, or a human tracking platform. Optionally, the data center is coupled to a plurality of data centers spatially located throughout a geographical region. Optionally, the data center is owned by a commercial company or a government entity.

(86) Optionally, the digital signal processing device includes a host interface to a switch device and a line interface to the light engine device.

(87) Optionally, the major substrate member includes an electrical interface to the system board member, an optical interface to the system board member, and a mechanical interface to the system board member. Optionally, the mechanical interface is thermally configured using an attachment device to the system board member using a thermal interface region coupled to the attachment device.

(88) While the above is a full description of the specific embodiments, various modifications, alternative constructions and equivalents may be used. Therefore, the above description and illustrations should not be taken as limiting the scope of the present invention which is defined by the appended claims.

Claims

1. An integrated circuit comprising: a silicon photonics substrate comprising a silicon-based material and comprising silicon photonics components formed in the silicon photonics substrate, the silicon photonics components configured to receive and transmit optical signals, the silicon photonics substrate comprising electrical connections; a transimpedance amplifier chip arranged on the silicon photonics substrate, the transimpedance amplifier chip comprising a silicon-germanium material that is different from the silicon-based material of the silicon photonics substrate, the transimpedance amplifier chip connected via the electrical connections in the silicon photonics substrate to at least one of the silicon photonics components configured to receive an optical signal, the transimpedance amplifier chip configured to process a received optical signal and output a processed signal to a digital signal processor; and a driver chip arranged on the silicon photonics substrate, the driver chip comprising CMOS material that is different from the silicon-germanium material of the transimpedance amplifier chip and from the silicon-based material of the silicon photonics substrate, the driver chip connected via the electrical connections in the silicon photonics substrate to drive at least one of the silicon photonics components configured to generate a coherent optical signal for transmission; a laser device arranged on the silicon photonics substrate wherein the laser device is configured to output light; a splitter configured to split the light from the laser device into a first portion and a second portion; a polarization beam splitter rotator configured to split TE-mode and TM-mode of the received optical signal and rotate the TM-mode to TE-mode; a receiver configured to receive outputs of the polarization beam splitter rotator, receive the first portion of the light from the laser device, and detect TE-mode and TM-mode signals of the received optical signal, wherein the detected TE-mode and TM-mode signals of the received optical signal are output to the transimpedance amplifier chip; and a transmitter configured to receive the second portion of the light from the laser device and to generate the coherent optical signal for transmission.

2. The integrated circuit of claim 1 wherein the electrical connections in the silicon photonics substrate are configured to connect the transimpedance amplifier chip and the driver chip to a printed circuit board.

3. The integrated circuit of claim 1 wherein the silicon photonics substrate comprises a plurality of waveguides comprising silicon related materials formed on a silicon-on-insulator substrate, and wherein the waveguides are configured to interconnect the silicon photonics components formed in the silicon photonics substrate.

4. The integrated circuit of claim 1 wherein the transmitter comprises: a modulator configured to

receive the second portion of the light from the laser device, to modulate the second portion of the light, and to output modulated signals; a polarization beam splitter combiner configured to combine the modulated signals; and a polarization independent semiconductor optical amplifier connected to the polarization beam splitter combiner and configured to output the coherent optical signal for transmission.

5. The integrated circuit of claim 4 wherein the driver chip is configured to provide control signals to the modulator to modulate the second portion of the light received from the laser device.

6. The integrated circuit of claim 1 wherein the laser device comprises a laser diode, a tunable filter, and couplers integrated into the silicon photonics substrate.

7. The integrated circuit of claim 6 wherein: the couplers are configured to couple the light from the laser device with a wavelength tuned by the tunable filter to the transmitter and the receiver.
